Examiner only

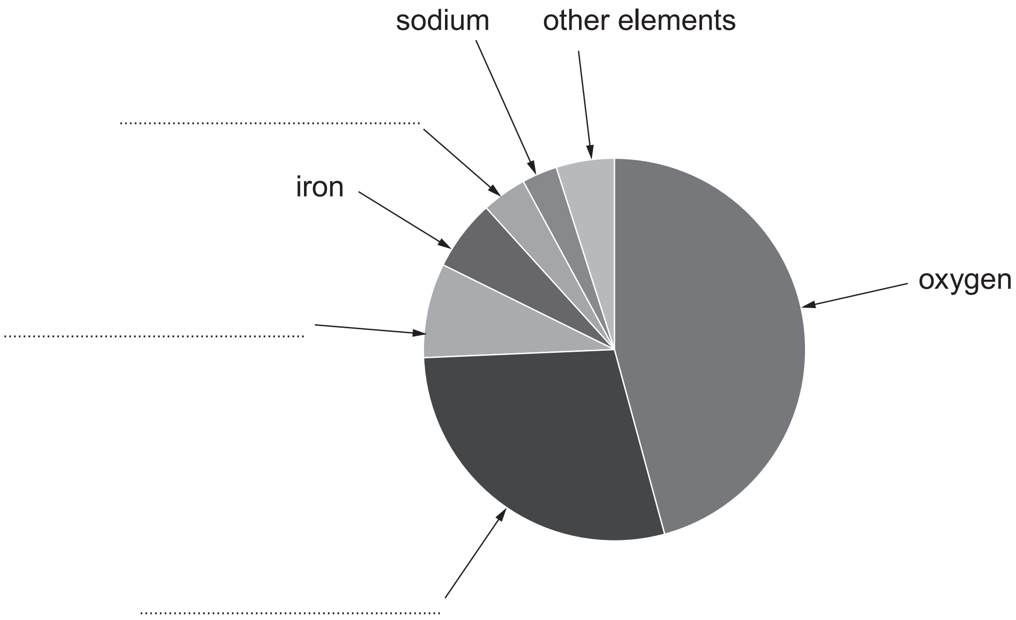
4. (a) The Earth's crust contains more than 80 elements, including many metals. Some of the metals, such as gold, are found 'native' or uncombined, but most are bonded with other elements to form compounds.

The crust can be thicker than 80 kilometres in some spots and less than one kilometre thick in others.

Just six elements make up almost all of the Earth's crust. These are aluminium, calcium, iron, oxygen, silicon and sodium.

Silicon is the second most abundant element. The amount of aluminium is approximately double the amount of calcium.

The chart shows the percentages of elements contained in the crust.



(i) Use the information given to **label** the remaining sections of the chart. [2]

(ii) Underline the approximate fraction of elements in the crust which are metals. [1]

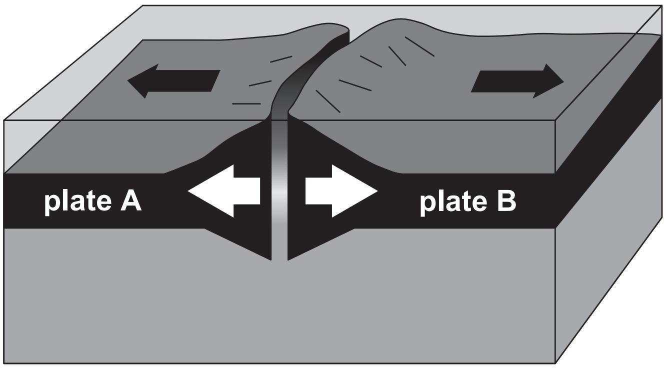
$\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{4}$ $\frac{3}{4}$

(iii) Tick (✓) the statement which best explains why most metals are **not** found native in the crust. [1]

most metals have higher melting points than gold	☐
most metals are magnetic	☐
most metals are more reactive than gold	☐
most metals are radioactive	☐

(b) Tectonic plates are huge sections of the crust that move very slowly above the mantle. The places where plates meet are called plate boundaries.

The diagram shows one type of plate boundary.



(i) Describe how new rock is formed at this plate boundary. [3]

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(ii) Underline the name of this type of plate boundary. [1]

conservative **destructive** **constructive**

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